

Remember that variable rendering supports dot notation. For the preceding to work, `instruments` must be iterable and contain objects that have `name` and `cost` attributes.

If the iterable given to `{% for %}` is empty, nothing in the block will be displayed. Depending on your situation, this might result in an empty list or table. You can either wrap the enclosing HTML in an `{% if %}` block or use the optional `{% empty %}` feature of the loop. Anything within the `{% empty %}` section of a `{% for %}` tag only gets rendered if there was nothing in the iterable being looped over:

```
<ul>
  {% for instrument in instruments %}
    <li> {{instrument.name}}: {{instrument.cost}}</li>
  {% empty %}
    <li> <i>No instruments found</i> </li>
  {% endfor %}
</ul>
```

In addition to the loop variable you define in `{% for %}`, there is an implicit variable as well: `forloop`. This is an object that contains attributes with information about the state of the loop as it is being iterated. The `forloop.counter` attribute is a 1-based count of the loop iteration. The Boolean values `forloop.first` and `forloop.last` indicate whether it is the first or last iteration of the loop, respectively. You can turn the previous example's `<ul>` into a sentence instead:

```
My favorite instruments are:
{% for instrument in instruments %}

  {% if forloop.last %} and {% endif %}

  {{ forloop.counter }}. {{instrument.name}}

  {% if forloop.last %}. {% else %}, {% endif %}
{% endfor %}
```

I've included extra newlines and indentation in this example to make the template readable—for HTML this wouldn't matter. If `instruments` contained two objects named `Bassoon` and `Oboe`, the result (stripped of newlines and extra spaces), would be

My favorite instruments are: 1. Bassoon, and 2. Oboe.

All the `forloop` attributes

The following is a complete list of the attributes available in the internal `forloop` object:

- `forloop.counter`—The iteration count of the loop, starting at 1
- `forloop.counter0`—The iteration count of the loop, starting at 0
- `forloop.revcounter`—The number of iterations remaining in the loop, with the last being 1
- `forloop.revcounter0`—The number of iterations remaining in the loop, with the last being 0